Routes without lifting the pencil

High school · graphs and logic · Progressive exercises · 06

Exercise 1. A graph has degrees 2, 3, 3, 4. How many edges does it have?

Exercise 2. Prove that the number of odd-degree vertices is even.

Exercise 3. Can a square and both diagonals be drawn without repeating an edge? Interior crossings are not vertices.

Exercise 4. Why do two disjoint cycles not admit one trail even though all degrees are even?